

Stanley College of Engineering & Technology for Women (A)**B.E (CSE/CME/AI&DS/IT) IV Semester Main Examination -July-2023****Operating Systems****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. List out different services of operating systems. [2M CO1 BTL1]
- b. What is a thread? Name the benefits of thread. [2M CO2 BTL1]
- c. Classify the necessary conditions for Deadlock. [2M CO3 BTL2]
- d. What is virtual memory? [2M CO4 BTL1]
- e. Describe page fault. [2M CO4 BTL2]
- f. Mention the key evaluation points of RAID System. [2M CO5 BTL1]

Part-B**4X12=48M**

2. a) Define OS and explain functions of OS. [6M CO1 BTL2]
- b) What is system call? Identify various types of system calls in detail. [6M CO1 BTL3]
3. a) From the below given data. Draw Gantt charts illustrating the execution of the processes using FCFS and SJF scheduling algorithms and calculate the average turnaround time and waiting time for them. [6M CO2 BTL2]

Processes	Arrival Time	Burst Time
P1	0	4
P2	1	2
P3	2	2
P4	3	5
P5	4	6

- b) Define process. Draw and explain various Process States in detail. [6M CO2 BTL3]
4. a) Illustrate about critical-section problem and Peterson's solution in concurrency. [6M CO3 BTL2]
- b) How to apply the monitors to solve the dining philosopher's problem with example? [6M CO3 BTL3]
5. a) Differentiate external fragmentation with internal fragmentation. [6M CO4 BTL4]
- b) Distinguish about Logical and Physical Addressing. [6M CO4 BTL4]
6. Consider the following page reference string:
 7,2,3,1,2,5,3,4,6,7,7,1,0,5,4,6,2,3,0,1
 Assuming the demand paging with three frames, determine how many page faults would occur for the following replacements algorithms?
 a. LRU b. FIFO c. Optimal [12M CO4 BTL5]
7. a) Summarize about file attributes, file operations, and file types. [6M CO5 BTL2]
- b) Discuss any two of disk scheduling algorithms with reference blocks. [6M CO5 BTL6]

Stanley College of Engineering & Technology for Women (A)**B.E (AI&DS) IV Semester (Main) Examinations-July-2023****Computer Networks****Time: 03 Hours****Max. Marks-60**

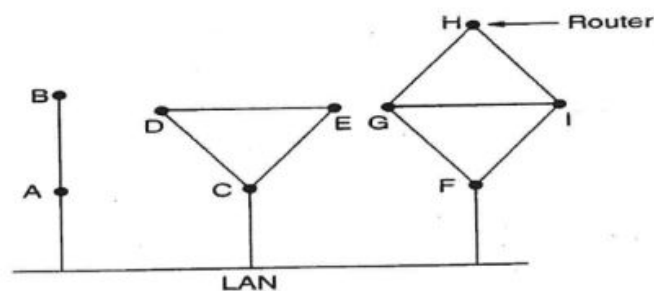
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4/33**Part-A****6X2=12M**

1. a. List the Topologies present in the computer networks. [2M CO1 BTL1]
- b. What is the meaning of single bit correction? [2M CO2 BTL1]
- c. Define Tunneling Mechanism. [2M CO3 BTL1]
- d. What is the meaning of Maximum Transmission unit? [2M CO4 BTL1]
- e. Define Domain Name Space. [2M CO5 BTL1]
- f. List any two error detection mechanism. [2M CO2 BTL1]

Part-B**4X12=48M**

2. a) Explain about OSI reference Model. [6M CO1 BTL2]
- b) Compare Circuit Switching and Packet Switching. [6M CO1 BTL4]
3. a) Demonstrate Hamming Code with example. [6M CO2 BTL2]
- b) Explain Go-back-N- ARQ protocol with neat diagram and example. [6M CO2 BTL5]
4. a) Compare Virtual and Datagram Networks. [6M CO3 BTL4]
- b) Apply Link State Routing for the given diagram. [6M CO3 BTL3]



5. a) Explain about crash recovery. [6M CO4 BTL2]
- b) Summarize about TCP connection establishment and release. [6M CO4 BTL2]
6. a) Compare SMTP, FTP and HTTP. [6M CO5 BTL4]
- b) Explain about Asynchronous and select system call. [6M CO5 BTL2]
7. a) Summarize about OSPF and BGP. [6M CO3 BTL2]
- b) Explain about UDP protocols. [6M CO4 BTL2]

Hall Ticket No. -

Code: R123743

Stanley College of Engineering & Technology for Women (A)

B.E (AI&DS) IV Semester (Main) Examinations-July-2023

Artificial Intelligence & Robotics

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
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Part-A

6X2=12M

1. a. Define concept of Agent based intelligent representation. [2M CO1 BTL1]
b. Write about Semantic nets. [2M CO2 BTL3]
c. Why do we use expert system in AI? [2M CO3 BTL1]
d. What is planning? [2M CO3 BTL1]
e. Where is Bayes rule used? [2M CO3 BTL3]
f. What are the future applications of Robot? [2M CO5 BTL4]

Part-B

4X12=48M

2. a) Explain the effect of overestimation and underestimation of it on A* algorithm. [6M CO1 BTL2]
b) Define Artificial Intelligence and list the task domains of Artificial Intelligence. [6M CO1 BTL1]
3. a) Describe the types of domains in Constraint Satisfaction Problems. [6M CO2 BTL2]
b) What are the challenges of Knowledge Representation and Reasoning (KRR)? [6M CO2 BTL2]
4. a) What are 4 applications of expert system? [6M CO3 BTL1]
b) Explain about partial order planning and hierarchical planning. [6M CO3 BTL2]
5. a) Explain about uncertainty and also describe about Belief networks. [6M CO3 BTL2]
b) Give detailed description of different types of fuzzy inference systems. [6 M CO3 BTL4]
6. a) Define the term 'Robot'. Explain different Robot types. [6 M CO4 BTL1]
b) Explain the basic components of Robot with suitable example. [6 M CO5 BTL2]
7. a) Describe different examples of adversarial search problems. [7M CO1 BTL2]
b) Discuss about fuzzy hedges. [5M CO3 BTL2]

Stanley College of Engineering & Technology for Women (A)
B.E (AI&DS) IV Semester (Main) Examinations-July-2023

Data Science

Time: 03 Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. Explain the concept of vectors in R programming. [2M CO4 BTL1]
b. Define database data and transactional data. [2M CO1 BTL1]
c. Explain data transformation with an example. [2M CO3 BTL2]
d. Explain the process of creating groupings in data analysis. [2M CO4 BTL2]
e. How do locate data on a web page before coding for web scraping. [2M CO4 BTL4]
f. What is the importance of string manipulation in data analysis? [2M CO4 BTL2]

Part-B

4X12=48M

2. a. Discuss the concept of control structures in R programming. [6M CO2 BTL2]
b. Explain the concept of UNIX time. Write R code to get what day of the week was '15th August 1947'. [6M CO2 BTL3]
3. a. Discuss Hypothesis Testing, Type-I and Type-II error. [6M CO2 BTL2]
b. Explain measuring data dissimilarity. How do you evaluate dissimilarity matrix for nominal and ordinal attributes? [6M CO3 BTL5]
4. a. What is merging and discuss the different types of merging with R code? [6M CO4 BTL3]
b. Explain the concept of NoSQL databases and their role in data storage. [6M CO1 BTL2]
5. a. How can you identify correlations and outliers in data analysis? [6M CO4 BTL4]
b. Discuss how box plots are used for identifying outliers and difference between histograms and bar plots. [6M CO3 BTL4]
6. a. Explain the concept of browser-based parsing in web scraping. [6M CO4 BTL2]
b. How can you perform screen reading using RSelenium in web scraping? [6M CO2 BTL3]
7. Explain the data science process in detail, highlighting the key steps involved and the role of each step in solving real-world problems. [12M CO5 BTL2]

Hall Ticket No. -

Code: R123748

Stanley College of Engineering & Technology for Women (A)

B.E (AI&DS) IV Semester (Main) Examinations-July-2023

Digital Electronics

Time: 03 Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. Prove that $ABC + ABC' + AB'C + A'BC = AB + AC + BC$ [2M CO1 BTL3]
b. Draw the logic diagram of full subtractor and explain. [2M CO2 BTL3]
c. Compare the PLD's: PAL and PLA. [2M CO3 BTL4]
d. Draw a serial in serial out shift register with initial state 1010 [2M CO4 BTL2]
e. Compare mealy and moore machine. [2M CO5 BTL4]
f. Explain the problem associated with SR latch. [2M CO4 BTL2]

Part-B

4X12=48M

2. a) Find all the prime implicants for the following Boolean function and determine which are essential: $F(A, B, C, D) = \sum m(2, 3, 4, 5, 6, 7, 9, 11, 12, 13)$ [6M CO1 BTL3]
b) Express the following Boolean expression in canonical product of sums form.
 $F(X, Y, Z) = (XY + Z)(XZ + Y)$ [6M CO1 BTL3]
3. a) Explain the working of 16 X 1 Multiplexer with a diagram. [6M CO2 BTL2]
b) Design a comparator using combinational gates. The comparator is 2-bit. Comparator which compares A(a1,a0) with B(b1,b0) and gives out 3-o/p's A=B, A<B, A>B [6M CO2 BTL3]
4. a) Implement the following two Boolean functions with a PLA: [6M CO3 BTL3]
 $F1(A, B, C) = \sum(0, 3, 5, 6)$
 $F2(A, B, C) = \sum(0, 1, 5, 7)$
b) Explain the structure of CPLDs with logic diagram [6M CO3 BTL2]
5. a) Draw the logic diagram of positive edge triggered S-R flip-flop and explain its operation with the help of truth table. [6M CO4 BTL2]

